UNIT 2 · TOPIC 8(A)

Cutting Principles and Cutter-Bar Design

Syllabus topic: Design of harvesters and threshers

Two modes of cutting

Mode	How it works	Used in
Shear (supported)	The stalk is cut between two edges — the moving knife section and the stationary ledger plate on the guard; works at moderate knife speeds.	Cutter-bar mowers, reapers, combine headers
Impact (free/inertia)	A single high-speed edge cuts the unsupported stalk, whose own inertia provides the reaction; needs a MINIMUM knife velocity (order of 20–50 m/s depending on crop). Below the critical speed the stalk is pushed over, not cut.	Rotary/disc mowers, flail harvesters, brush cutters

Cutter-bar construction

The knife (sickle) is made of knife sections riveted to a knife back. It reciprocates inside guards fitted with ledger plates; wearing plates support the knife back; knife clips hold the sections against the ledger plates. A pitman (connecting rod) drives it from a crank, or a modern wobble/swash-plate drive.

- **Standard dimensions:** knife-section pitch = guard spacing = stroke = **76.2 mm (3 in)**.
- Crank speed typically **800–1000 rpm** → **1600–2000 cutting strokes per minute** (each crank revolution = 2 cutting strokes: one left, one right).

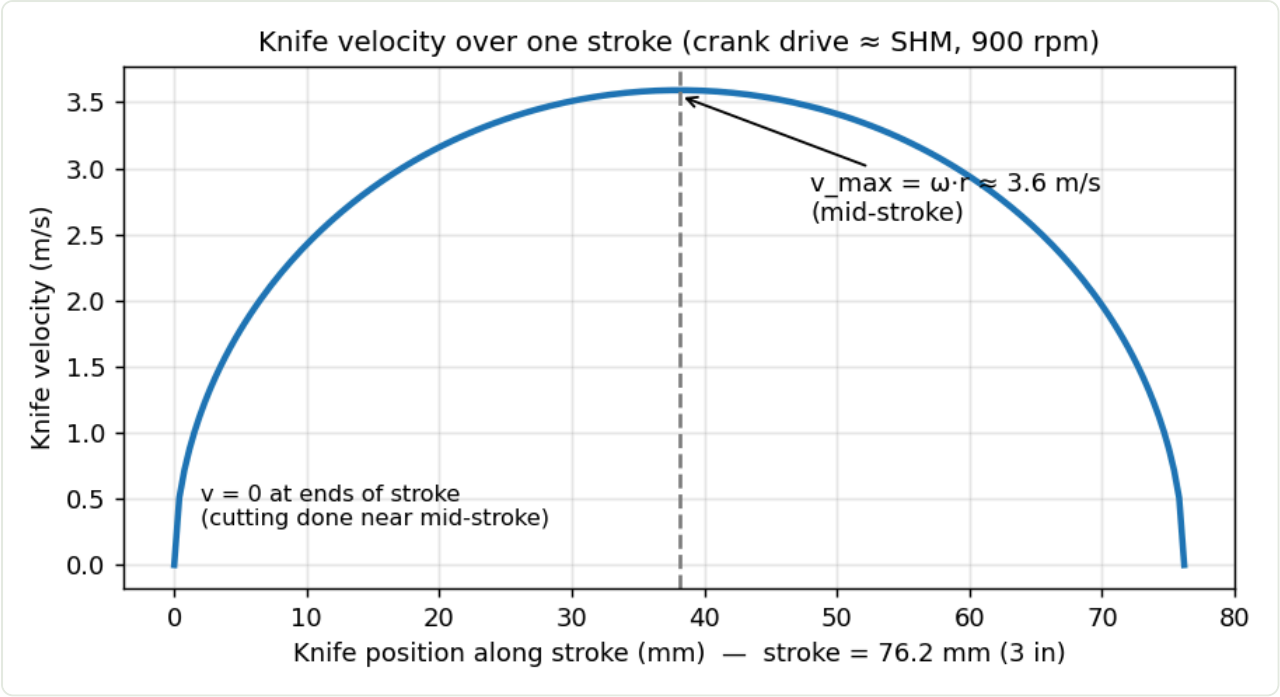
Knife kinematics

The crank–pitman drive gives approximately simple harmonic motion (r = half-stroke):

$$x = r\cdot\cos(\omega t);\ v = \omega\cdot\sqrt{(r^2 - x^2)};\ v_max = \omega\cdot r$$

Velocity is **zero at both ends** of the stroke and **maximum at mid-stroke** — so design cutting to happen in the middle of the stroke, where knife speed is high.

Example. 3-in (76.2 mm) stroke, crank 900 rpm → ω = 2π × 900/60 = 94.2 rad/s; r = 38.1 mm → v_max = 94.2 × 0.0381 ≈ **3.6 m/s**. Average knife speed = 2 × stroke × rpm/60 = 2 × 0.0762 × 15 ≈ **2.3 m/s**.



Knife velocity over one stroke

Register, alignment and lead

- **Register:** the knife is in proper register when each section centres exactly on its guard at **each end** of the stroke. Adjust it by shifting the cutter bar (or the pitman length).
- **Alignment & lead:** in correct alignment the pitman and knife are in line at working position. Set the outer end of the cutter bar to **lead** (run ahead of) the inner end by roughly **1–2 cm per metre** of bar length, so the bar's rearward flexing under cutting load brings it truly perpendicular to travel and preserves register.
- **Serrations** on ledger plates/knife edges increase friction so stalks are gripped and cut instead of slipping forward off the edges. The included angle between knife and ledger edges must not exceed the sum of the two friction angles, or the stalk slips.

Cutting pattern and feed

$$\text{Feed (forward travel per stroke)} = v_{\text{forward}} \times \text{time per stroke}$$

- With a crank at 880 rpm and 5 mph forward speed, feed ≈ **3 in per stroke**. Stalks in part of the cut area must then be crowded/deflected before being cut.
- Excessive feed causes bunching, ragged stubble and higher stubble loss.
- A short stroke and correct guard spacing minimise side deflection of stalks.

Practice MCQs

Q146. In a conventional mower cutter bar, knife-section pitch, guard spacing and stroke are all equal to:

(a) 50.8 mm
(b) 76.2 mm
(c) 101.6 mm
(d) 152.4 mm

Q147. Typical mower crank speed is 800–1000 rpm; the corresponding cutting strokes per minute are:

(a) 400–500 (b) 800–1000
(c) 1600–2000 (d) 3200–4000

Q148. Knife velocity in a crank-driven cutter bar is maximum at:

(a) Start of stroke (b) End of stroke
(c) Mid-stroke (d) Uniform throughout

Q149. For 76.2-mm stroke at 900 rpm crank speed, peak knife velocity is about:

(a) 1.2 m/s (b) 2.3 m/s
(c) 3.6 m/s (d) 7.2 m/s

Q150. A knife is in proper REGISTER when knife sections centre on the guards:

(a) At mid-stroke (b) At each end of the stroke
(c) Only while cutting (d) At maximum velocity

Q151. 'Lead' given to the outer end of a cutter bar compensates for:

(a) Wheel slip (b) Rearward flexing of the bar under cutting load
(c) Crop moisture (d) Pitman wear

Q152. Serrations on ledger plates mainly serve to:

(a) Reduce weight (b) Increase friction so stalks do not slip off the edges
(c) Sharpen the knife (d) Reduce noise

Q153. Rotary (disc) mowers cut by:

(a) Shear between two edges (b) Impact/free cutting using stalk inertia
(c) Tearing (d) Sawing

Q154. Impact cutting is possible only above a:

(a) Minimum stalk diameter (b) Minimum knife velocity
(c) Maximum moisture (d) Minimum stubble height

Answers

Q146 — (b) 76.2 mm

Q147 — (c) 1600–2000 — *Each crank revolution gives two cutting strokes (left + right), so 800–1000 rpm → 1600–2000 strokes/min.*

Q148 — (c) Mid-stroke

Q149 — (c) 3.6 m/s — *$\omega = 2\pi \times 900/60 = 94.2$ rad/s; $r = 38.1$ mm → $v_{max} = \omega r \approx 3.6$ m/s. Velocity is zero at stroke ends — hence cutting is timed near mid-stroke.*

Q150 — (b) At each end of the stroke

Q151 — (b) Rearward flexing of the bar under cutting load

Q152 — (b) Increase friction so stalks do not slip off the edges

Q153 — (b) Impact/free cutting using stalk inertia

Q154 — (b) Minimum knife velocity

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 14 (Hay Harvesting: Mowing) and Ch. 17 (Grain and Seed Harvesting); Jagdishwar Sahay — Elements of Agricultural Engineering; BIS thresher standards (IS 9129 safety); NPTEL Farm Machinery lectures.